

Lifelong Adaptation and Precision Modification: A Comprehensive Survey on Continual Learning and Model Editing for Foundation Models

1 Introduction

The paradigm of foundation models has fundamentally shifted artificial intelligence from task-specific specialization to general-purpose adaptability, yet the prevailing training methodology remains predominantly static. Traditional approaches rely on a one-time pre-training phase followed by limited fine-tuning, creating systems that are effectively frozen in time despite operating in rapidly evolving environments [1]. This static nature poses a critical limitation: as world knowledge changes and data distributions shift non-stationarily, models risk becoming obsolete or propagating outdated information. Consequently, there is an urgent necessity to transition toward dynamic adaptation mechanisms that enable lifelong learning, allowing foundation models to continuously accumulate knowledge without catastrophic forgetting [2].

The scope of this adaptation encompasses two distinct yet complementary problem spaces: continual learning and model editing. Continual learning addresses the broad acquisition of new capabilities through sequential task exposure, aiming to maintain stability on historical tasks while preserving plasticity for new ones [3]. In contrast, model editing focuses on the precise, localized correction of specific facts or behaviors, such as updating outdated entities or mitigating biases, without requiring global retraining [4]. While continual learning strategies often leverage parameter-efficient tuning modules like adapters or prompts to isolate new knowledge [5], model editing techniques frequently employ locate-and-edit methodologies or external memory structures to inject facts directly [6]. However, treating these fields in isolation fails to address the complex reality of deployed systems, which require both broad domain adaptation and surgical knowledge correction simultaneously.

Central to this unified challenge is the “Stability-Plasticity-Locality” trilemma. In over-parameterized regimes, increasing plasticity to accommodate new tasks often degrades stability, leading to significant forgetting of pre-trained capabilities [7]. Conversely, rigorous constraints to preserve stability can render the model intransigent to new information. Model editing introduces a third dimension: locality. Precise updates must alter specific outputs without inducing unintended side effects on unrelated knowledge, a constraint often violated by naive fine-tuning which causes widespread representation shifts [8]. Recent theoretical work suggests that achieving high performance across all three dimensions simultaneously is theoretically bounded, creating an “impossible triangle” where optimizing for one metric inherently compromises the others in lifelong scenarios [9].

Emerging trends indicate a move toward hybrid architectures that decouple static pre-trained weights from dynamic update modules. Approaches such as discrete key-value adapters allow for thousands of sequential edits by writing to a latent codebook rather than modifying base weights, thereby preserving the integrity of the original model [10]. Similarly, dual-memory schemes separate long-term pre-trained knowledge from short-term edited facts, utilizing routing mechanisms to reconcile potential conflicts [9]. Despite these advances, significant challenges remain in managing interference during infinite-horizon adaptation and ensuring logical consistency when new edits contradict existing knowledge [11]. This survey aims to bridge these historically separate domains, proposing a unified taxonomy and analyzing the synergistic potential of integrating continual learning and model editing to realize truly self-evolving foundation models.

2 Conceptual Frameworks and Problem Formalizations

2.1 Unified Taxonomy of Adaptation Scenarios and Granularity

To unify the historically disjoint fields of continual learning (CL) and model editing, we propose a hierarchical taxonomy grounded in the nature of data streams, the specificity of knowledge updates, and the granularity of intervention. Traditional CL formulations often rely on rigid task boundaries, categorizing scenarios into task-incremental, class-incremental, and domain-incremental settings [12; 13]. However, foundation models operate in more fluid environments where concepts recur and distributions shift non-stationarily. We thus extend this spectrum to include “Class-Incremental with Repetition” and online streaming settings with blurry boundaries, which better reflect real-world deployment where data arrives as a continuous flow rather than discrete episodes [14; 15]. In contrast, model editing scenarios are defined not by task sequences but by the intent of modification: factual updating, bias mitigation, or error rectification [16; 17]. This distinction necessitates a shift from minimizing catastrophic forgetting across tasks to maximizing edit locality and portability while preventing interference [18; 19].

The granularity of adaptation serves as the second axis of our taxonomy, ranging from coarse-grained behavioral alignment to microscopic parameter localization. At the coarsest level, continual pre-training and instruction tuning involve global updates to expand core knowledge or align behaviors, often leveraging parameter-efficient modules like adapters or prompts to isolate plasticity [20; 5]. Moving finer, model editing targets specific factual associations, typically localized to mid-layer feed-forward networks, allowing for precise weight modifications without global retraining [18; 21]. The most granular interventions involve neuron-level attribution or activation steering, where individual computational units are patched or hidden states are manipulated during inference to correct specific errors [22; 23]. Emerging hybrid approaches, such as discrete key-value adapters, bridge these granularities by writing edits into a latent codebook, enabling thousands of sequential modifications without altering the base weights [10; 9].

Temporal dynamics further differentiate these scenarios, imposing distinct constraints on latency and consistency. Offline sequential batches allow for rehearsal and consolidation strategies like generative replay or gradient coreset selection [24; 25]. Conversely, online streaming and post-deployment editing demand low-latency updates, often precluding extensive replay and necessitating closed-form solutions or retrieval-augmented mechanisms [4; 6]. A critical challenge in this unified view is the “impossible triangle” of reliability, generalization, and locality, which becomes increasingly difficult to satisfy as the number of sequential edits grows due to knowledge superposition [9; 11]. Future research must therefore transcend isolated evaluations, developing frameworks that dynamically adjust adaptation granularity based on the temporal urgency and semantic scope of the incoming data stream, effectively treating foundation models as configurable systems capable of both broad evolution and precise surgical correction [26; 27].

2.2 Mathematical Formalizations of Stability, Plasticity, and Locality

Building on the hierarchical taxonomy and temporal dynamics established previously, this subsection formalizes the competing objectives of stability, plasticity, and locality as a constrained optimization landscape. In continual learning (CL), the objective is traditionally framed as minimizing the loss on a current task t , denoted $\mathcal{L}_t(\theta)$, subject to inequality constraints that bound performance degradation on previous tasks $\tau < t$. This trade-off can be modeled via Lagrangian duality, where stability is enforced by penalizing deviations from optimal parameters of past tasks, often approximated through Fisher information matrices or gradient projections [28; 29]. However,

theoretical analysis suggests that optimal CL algorithms avoiding catastrophic forgetting entirely require solving NP-hard problems and retaining perfect memory, implying that practical methods must rely on approximations such as episodic memory buffers or generative replay to manage the stability-plasticity dilemma [30; 31]. Recent advances refine this by maximizing transfer while minimizing interference through gradient alignment, effectively optimizing a temporally symmetric objective that accounts for future task distributions [32].

In the domain of model editing, the formalization shifts from sequential task optimization to precise parameter manipulation, reflecting the intent-driven scenarios outlined in our taxonomy. The core objective is defined by a preservation-memorization function, where the goal is to inject a new fact (s, r, o) into the model weights θ such that the output probability $P_\theta(o|s, r)$ is maximized, while minimizing the KL-divergence between the pre-edit and post-edit distributions for unrelated inputs [33]. Metrics such as Edit Success Rate (ESR), Locality, and Portability quantify this balance, yet theoretical bounds indicate that uniformly trainable models without distal interference necessitate exponential capacity, rendering strict locality impossible in polynomial-complexity architectures without architectural constraints [4; 6]. The phenomenon of knowledge superposition in over-parameterized transformers introduces an unavoidable interference term; as the number of sequential edits increases, the condition number of the edited weight matrices can degrade, leading to “model collapse” unless specific orthogonality conditions or regularization terms are maintained [26; 11].

Furthermore, the distinction between editing long-term memory (parameters) and working memory (activations) reveals an “impossible triangle” where reliability, generalization, and locality cannot be simultaneously optimized in lifelong settings without dual-memory architectures [9]. Theoretical investigations into linear regression settings provide exact bounds on forgetting, demonstrating that cyclic task presentation yields forgetting rates scaling with T^2 , whereas random ordering mitigates this effect, offering insights into the geometry of loss landscapes in high-dimensional spaces [34]. Ultimately, bridging these fields requires viewing adaptation not merely as weight updates but as navigating a complex manifold where plasticity is preserved through mechanisms like synaptic scaling or regenerative regularization, ensuring that the model remains capable of infinite-horizon learning without succumbing to representational collapse [35; 36]. These theoretical imperatives directly motivate the architectural strategies discussed next, which propose dynamic capacity allocation and modular “bricks” to navigate the non-stationary nature of both data streams and knowledge graphs [9].

2.3 Lifecycle Stages and Modular Adaptation Architectures

The adaptation of foundation models is best conceptualized not as a monolithic retraining event, but as a continuum of distinct lifecycle stages, each demanding specific architectural interventions to balance stability, plasticity, and computational efficiency. This subsection delineates these phases—continual pre-training, instruction tuning, and post-deployment editing—and proposes a modular architecture where models function as configurable assemblies of functional “bricks.”

In the earliest stage, **Continual Pre-training**, the objective is the expansion of core knowledge representations through exposure to massive, evolving corpora. Naive fine-tuning in this regime often leads to catastrophic forgetting, particularly as model scale increases [7]. To mitigate this, strategies must shift from static architectures to dynamic capacity expansion. Approaches like [37] demonstrate that integrating Mixture-of-Experts (MoE) structures allows models to dynamically allocate new experts for emerging domains while regularizing pretrained weights, thereby preserv-

ing generalizability. Similarly, [20] highlights that distillation-based consolidation is critical for retaining downstream performance across temporally shifting data distributions. Theoretical insights suggest that pre-trained weights naturally reside in wider loss basins, which can be explicitly optimized to further alleviate forgetting during sequential domain adaptation [38].

The intermediate phase, **Parameter-Efficient Instruction Tuning**, focuses on behavioral alignment and task-specific adaptation without altering the foundational backbone. Here, the paradigm shifts toward freezing the core model and learning lightweight, task-specific modules. This modular isolation is exemplified by Low-Rank Adaptation (LoRA), where recent findings reveal an asymmetry in matrix roles; specifically, optimizing only the projection matrix B while keeping A fixed can enhance generalization bounds and reduce interference [39]. Furthermore, frameworks like [5] propose accumulating task-specific knowledge into offline modules via momentum updates, effectively treating adapters as retrievable bricks that can be ensembled during inference. This approach aligns with “Slow Learner with Classifier Alignment” [40], which advocates for minimal backbone updates to prevent progressive overfitting, instead relying on classifier realignment to accommodate new tasks.

The final stage, **Post-Deployment Editing**, necessitates immediate, localized corrections for factual errors or emerging knowledge gaps. This phase treats the model as a mutable system capable of “growing” or “patching” specific functional units. Direct weight modification methods, such as [4], utilize gradient decomposition to apply local edits, though they often struggle with sequential interference. Consequently, architectural solutions that decouple memory from computation are gaining prominence. For instance, [10] introduces discrete codebooks in the latent space to store thousands of sequential edits without modifying base weights, effectively creating a side-memory for dynamic updates. Similarly, [9] proposes a dual parametric memory scheme with a router to direct queries to either pretrained or edited subspaces, resolving the trade-off between reliability and locality. Extending this modularity, [22] demonstrates that adding and training negligible numbers of neurons can rectify errors sequentially, embodying the concept of neural patching.

Synthesizing these stages reveals a trajectory toward fully modular foundation models where adaptation is achieved through the dynamic retrieval, merging, and expansion of functional components rather than global parameter updates. Future architectures must increasingly integrate these disjoint strategies—combining the capacity scaling of continual pre-training, the isolation of parameter-efficient tuning, and the precision of memory-augmented editing—to realize robust, lifelong learning systems capable of evolving indefinitely without performance degradation.

2.4 Theoretical Underpinnings of Forgetting in Over-Parameterized Regimes

While the preceding section conceptualized foundation model adaptation as a modular lifecycle of pre-training, tuning, and editing, this subsection grounds those architectural strategies in the unique theoretical landscape of over-parameterized regimes. The transition from small-scale networks to foundation models fundamentally alters the mechanics of catastrophic forgetting. In classical regimes, forgetting is often attributed to the violent displacement of decision boundaries by new task gradients. However, in over-parameterized regimes, the geometry of the loss landscape and the dynamics of gradient descent exhibit distinct behaviors that challenge these traditional intuitions. Recent theoretical analyses suggest that while increased model width can initially mitigate forgetting through wider loss basins, it may paradoxically exacerbate interference as scale grows, particularly when task distributions diverge significantly [38; 7]. This non-monotonic relationship implies that simply scaling parameters does not guarantee robustness against distribution shifts;

rather, the interaction between task similarity and representational capacity dictates the trajectory of knowledge retention.

A critical theoretical insight arises from the study of over-parameterized linear models, which provide exact expressions for expected forgetting. These models reveal that forgetting is not merely a function of memory capacity but is deeply tied to the ordering of tasks and the spectral properties of input data distributions [34; 41]. Specifically, in cyclic task presentations, forgetting bounds scale with the square of the number of tasks, whereas random ordering can lift this constraint, suggesting that the temporal structure of data streams interacts non-trivially with high-dimensional geometry. Furthermore, the phenomenon of “lazy training” in wide networks, where parameters remain close to their initialization, influences gradient orthogonality. While this regime can preserve prior representations by restricting updates to low-rank subspaces, it also limits the model’s ability to encode distinct features for highly dissimilar tasks, leading to a form of representational collapse where shared resources become saturated [42; 43]. This saturation directly motivates the parameter-efficient isolation strategies discussed in the instruction tuning phase, where freezing the backbone prevents the degradation of these shared subspaces.

The concept of knowledge superposition further complicates the theoretical understanding of editing and continual learning in these regimes, providing the mathematical justification for the “neural patching” and dual-memory schemes described in the post-deployment stage. In foundation models, multiple facts are encoded in overlapping subspaces of the same neurons, creating an unavoidable interference term that scales with the number of sequential modifications unless strict orthogonality conditions are met [11; 44]. This superposition creates a “reversal curse” and limits the portability of edits, as modifying a specific fact can inadvertently perturb the condition number of the weight matrices, degrading general abilities across unrelated domains [45; 46]. Theoretical bounds indicate that uniformly trainable models without distal interference require exponential capacity, formalizing the necessity of architectural constraints—such as discrete key-value adaptors or dual-memory schemes—to achieve polynomial-complexity lifelong adaptation [10; 9].

Moreover, the loss of plasticity presents a distinct theoretical challenge separate from forgetting, underscoring the need for the dynamic capacity expansion highlighted in the continual pre-training phase. Over-parameterized models subjected to long sequences of non-stationary updates tend to converge to sharp minima or saturate activation patterns, rendering them unable to learn new concepts even if past knowledge is preserved [47]. This suggests that the stability-plasticity dilemma in foundation models is not just about protecting old weights but about maintaining the spectral properties of the Hessian that allow for future learnability. Future theoretical work must therefore move beyond static bounds on forgetting to dynamic analyses of how gradient decomposition and layer-wise optimization can decouple task-specific interference from shared representational updates, ensuring that the “impossible triangle” of reliability, generalization, and locality can be approximated in infinite-horizon learning scenarios [48; 49].

3 Algorithmic Strategies for Continual Foundation Model Learning

3.1 Parameter-Efficient Adaptation and Modular Isolation

Parameter-efficient adaptation has emerged as a foundational strategy for continual learning in foundation models, addressing the critical tension between stability and plasticity by decoupling new task acquisition from the immutable core weights. Rather than updating the entire parameter space, these methods isolate plasticity within lightweight, trainable modules—such as adapters, low-rank matrices, or prompt vectors—thereby preserving the rich, generalizable representations encoded

during pre-training. This architectural isolation fundamentally alters the forgetting dynamics; by freezing the backbone, the model prevents the overwriting of previously learned features, a phenomenon often exacerbated by full fine-tuning in non-stationary environments.

The evolution of this paradigm has progressed from static, task-specific module allocation to dynamic, self-expanding architectures capable of automatic capacity scaling. Early approaches utilized fixed adapter configurations or prompt pools, where distinct modules were assigned to identified tasks. However, such methods often struggle in task-free or highly dynamic streams where task boundaries are ambiguous. To address this, recent frameworks like [50] introduce attention-based mechanisms that assemble input-conditioned prompts end-to-end, significantly enhancing plasticity without sacrificing the rehearsal-free benefits of prompting. Similarly, [51] demonstrates that sequentially concatenating learned soft prompts allows for effective knowledge accumulation and forward transfer, outperforming state-of-the-art methods by maintaining a growing context of task-specific cues while keeping the base model frozen.

A critical advancement in this domain is the recognition of asymmetry within low-rank adaptation mechanisms. While Low-Rank Adaptation (LoRA) is widely adopted for its efficiency, recent theoretical and empirical analyses reveal that the constituent matrices play distinct roles: one extracts features while the other projects them. [39] exploits this insight, showing that optimizing only the projection matrix while keeping the feature extractor fixed can yield superior generalization bounds and reduced forgetting compared to updating both. Building on this, [52] proposes an interference-free subspace design, ensuring that the parameter updates for new tasks are orthogonal to the subspaces occupied by previous tasks, thereby mathematically guaranteeing the elimination of inter-task interference.

Furthermore, the integration of Mixture-of-Experts (MoE) structures has enabled modular expansion that scales sub-linearly with the number of tasks. [37] illustrates how dynamically adding specialized experts allows models to adapt to distribution shifts while regularizing the pre-trained core. In the vision-language domain, [53] combines MoE adapters with a distribution-discriminative router to automatically direct in-distribution and out-of-distribution inputs to appropriate processing paths, preserving zero-shot capabilities while acquiring new skills. This modularity extends to the concept of “configurable foundation models,” where functional “bricks” can be retrieved, merged, or grown [54], treating the model as a composable system rather than a monolithic entity.

Despite these advances, challenges remain in managing the growing memory footprint of accumulated modules and ensuring efficient inference. Strategies like [55] attempt to unbind tuners from the backbone to facilitate multi-task inference with a single forward pass, while [56] utilizes semantic-guided prototype complementation to synthesize old class features, mitigating the need for storing extensive historical data. Future research must further refine the trade-off between module granularity and computational overhead, potentially leveraging neuro-inspired sparse connectivity [57] to create ever-more efficient, lifelong adaptive systems that mimic the brain’s ability to grow and rewire without catastrophic disruption.

3.2 Generative Replay and Latent Space Consolidation

While Parameter-Efficient Adaptation (PEA) addresses forgetting by structurally isolating plasticity within lightweight modules, Rehearsal-based strategies tackle the same challenge by explicitly preserving the statistical characteristics of past distributions. Rather than modifying the model architecture, these methods interleave current training streams with synthesized or compressed representations of historical data, directly countering the distributional shift that drives catastrophic

forgetting. Traditional generative replay mechanisms, inspired by hippocampal consolidation, employ a dual-model architecture where a generator synthesizes historical data to train a solver [58]. However, while effective in low-dimensional domains, pixel-level generation faces significant scalability barriers and privacy constraints when applied to foundation models, where raw data retention is often prohibited. Consequently, recent advancements have pivoted toward latent space consolidation, operating within the high-dimensional embedding manifolds of pre-trained backbones to achieve memory efficiency without sacrificing fidelity.

A primary evolution in this domain is the transition from input-space replay to latent feature matching. Methods such as Latent Replay store activation volumes at intermediate layers rather than raw inputs, significantly reducing computational overhead while maintaining representational stability [59]. This approach is particularly critical for edge deployment, where quantized latent replays enable continual adaptation on resource-constrained hardware [60]. To further enhance the quality of synthesized knowledge, advanced frameworks align local generative models with global latent representations, ensuring that new task embeddings remain consistent with the manifold of prior experiences [61]. Moreover, diffusion models have been leveraged to guide the rehearsal process, generating samples specifically targeting regions of the feature space most prone to forgetting, thereby outperforming random sampling strategies [62].

Complementing these generative approaches, prototype condensation techniques offer a highly parameter-efficient alternative by compressing historical class distributions into single, high-density representative vectors. Unlike exemplar-based methods that suffer from recency bias, prototype evolution allows for the dynamic updating of class centers in a shared latent space, facilitating robust online learning from non-stationary streams [15]. By employing supervised contrastive losses, these methods enforce tight clustering of intra-class samples while maximizing inter-class separation, effectively mitigating the representational overlap that typically drives forgetting [63]. Recent innovations such as Elastic Feature Consolidation extend this paradigm to cold-start scenarios, utilizing second-order approximations of feature drift to regularize prototype updates and preserve plasticity [64]. Furthermore, steering prototypes via prompt-tuning enables rehearsal-free adaptation in transformer architectures, narrowing the performance gap with joint training [65].

Despite these advances, challenges remain regarding the stability of generative models in sequential settings and the potential for “model collapse” when training on synthetic data loops [66]. Future directions must therefore integrate rigorous theoretical bounds on latent space drift with hybrid architectures that combine generative replay with retrieval-augmented mechanisms. Such synergies will be essential for scaling continual learning to the infinite horizons required by autonomous foundation agents, ensuring that knowledge consolidation remains both computationally tractable and semantically precise. While these data-centric and latent-space methods focus on consolidating knowledge through representation, the following section explores an alternative paradigm that imposes mathematical constraints directly on the optimization landscape to enforce stability without relying on external memory or structural expansion.

3.3 Regularization Constraints and Plasticity Preservation

Regularization-based strategies constitute a foundational pillar in continual foundation model learning, operating by imposing mathematical constraints on parameter updates to enforce stability while preserving plasticity. Unlike architectural expansion or replay mechanisms, these methods modulate the optimization landscape directly, penalizing deviations from critical states identified in previous tasks. The seminal approach of Elastic Weight Consolidation and its variants relies on es-

timating parameter importance, typically via the Fisher Information Matrix, to construct quadratic penalties that restrict updates to directions orthogonal to previously learned decision boundaries [67; 28]. However, in the over-parameterized regimes of foundation models, precise curvature estimation becomes computationally prohibitive, prompting the development of approximations such as the extended Kronecker-factored approximate curvature, which accounts for inter-example dependencies induced by batch normalization layers [68].

A distinct line of inquiry shifts focus from parameter-space regularization to functional and gradient-space constraints. Orthogonal Gradient Descent (OGD) and its successors, such as Adam-NSCL, project candidate updates onto the null space of previous task gradients, theoretically guaranteeing zero forgetting under linear assumptions [69; 29]. Advanced variants like Class Gradient Projection refine this by calculating subspaces at the class level rather than the task level, mitigating interference caused by class deviation within tasks [70]. Similarly, functional regularization approaches constrain the model’s output distribution rather than its weights, often utilizing memorable past examples to define a functional prior that preserves decision boundaries without storing raw data [71]. Recent theoretical work further suggests that projecting gradients onto the residual space of pre-trained subspaces can minimize interference while facilitating positive forward transfer in multimodal instruction tuning [72].

Despite their efficacy in mitigating catastrophic forgetting, traditional regularization methods often inadvertently induce a “loss of plasticity,” where the model progressively loses its capacity to learn new concepts due to the accumulation of restrictive penalties and the saturation of neural units [47]. Addressing this dual challenge requires regenerative regularization techniques that prevent weights from converging to static configurations. For instance, L2-regularization toward the initial weight distribution, rather than the origin, has been shown to maintain trainability over infinite horizons by preserving the spectral properties of weight matrices [73]. Furthermore, spectral regularization strategies that explicitly constrain singular values ensure gradient diversity is maintained, countering the rank collapse that typically hampers deep networks in sequential settings [73].

Emerging trends increasingly integrate uncertainty quantification to dynamically balance these competing objectives. Uncertainty-regularized Continual Learning (UCL) introduces node-wise uncertainty to adaptively freeze important parameters while allowing active learning in others, effectively solving the trade-off between stability and plasticity without excessive memory overhead [28]. Similarly, entropy-based criteria can dynamically determine layer-wise plasticity factors, enabling the model to selectively freeze rigid layers while keeping others adaptable [74]. Future directions point toward hybrid frameworks that combine asymmetric regularization losses with distribution-aware calibration, ensuring that foundation models retain their zero-shot capabilities while adapting to non-stationary domains [75; 76]. Ultimately, the evolution of regularization constraints lies in moving from static penalties to dynamic, context-aware mechanisms that treat plasticity not as a byproduct, but as a conserved resource essential for lifelong adaptation.

3.4 Hybrid Architectural Strategies and Neural Patching

Building upon the dynamic, context-aware regularization mechanisms discussed previously, hybrid architectural strategies represent a further paradigm shift, moving beyond the dichotomy of static fine-tuning and episodic replay toward mutable, modular systems capable of localized repair and dynamic expansion. While regularization methods treat plasticity as a conserved resource within a fixed parameter space, these hybrid frameworks address the “stability-plasticity-locality” trilemma by re-conceptualizing foundation models not as monolithic entities, but as configurable assemblies

of functional components that can be incrementally updated to handle concept drift and specific failure modes. A primary manifestation of this approach is neural patching, where small auxiliary networks or specific neuron subsets are trained to correct misclassifications or backdoor vulnerabilities in a frozen backbone, effectively remediating errors without altering original weights. For instance, [22] demonstrates that adding and training a few neurons in the final feed-forward layer can successively correct thousands of errors while maintaining high locality and generality, outperforming traditional hypernetwork-based methods in sequential editing scenarios. Similarly, [77] introduces weight interpolation techniques to expand task coverage in open-vocabulary models like CLIP, achieving significant accuracy gains on specific tasks while preserving zero-shot performance on others.

Complementing localized patching, retrieval-augmented continuous prompting externalizes memory to handle the scalability limits of parameter-based updates, offering a distinct alternative to the gradient-projection and spectral constraints of prior sections. Instead of modifying weights, these systems convert retrieved facts into continuous prompts that guide model behavior dynamically. [78] proposes RECIPE, a framework that integrates a knowledge sentinel to determine retrieval relevance, converting knowledge statements into informative prompts that refine responses without catastrophic forgetting. This decoupling of knowledge storage from reasoning capabilities allows for million-scale knowledge injections, as seen in architectures like [6], which utilizes explicit memory stores to modulate base model predictions, thereby overcoming the expressiveness limitations of direct weight modification. Furthermore, the concept of configurable foundation models decomposes large networks into functional “bricks”—emergent or customized neuron partitions—that support operations such as dynamic retrieval, merging, and growing [54]. This modular perspective enables the model to “grow” new capabilities by instantiating new bricks or routing inputs to specialized experts, mirroring the efficiency of biological neuroplasticity and extending the adaptive potential initiated by regenerative regularization.

Despite their promise, hybrid strategies face critical challenges regarding interference management and architectural complexity that differ from the plasticity saturation issues of pure regularization methods. The integration of multiple adaptation mechanisms introduces potential conflicts between patched modules and retrieved contexts, necessitating sophisticated routing algorithms to ensure logical consistency. While [79] highlights the benefits of composing atomic skill modules, it also underscores the computational overhead of searching optimal module combinations in long task streams. Moreover, the theoretical bounds of such hybrid systems remain under-explored, particularly concerning the accumulation of errors when patching and retrieval mechanisms interact over infinite horizons. Future research must focus on developing unified optimization objectives that jointly train patching modules, retrieval encoders, and routing policies to minimize distal interference. By synthesizing architectural expansion with precise neural patching, these hybrid frameworks offer a robust pathway toward truly lifelong adaptive systems that maintain the integrity of pre-trained knowledge while seamlessly integrating new information, setting the stage for the advanced model editing techniques explored in the subsequent section.

4 Mechanisms for Precise Model Editing and Knowledge Correction

4.1 Causal Localization and Mechanistic Attribution

Causal localization and mechanistic attribution constitute the prerequisite phase for precise model editing, aiming to isolate the specific neural substrates responsible for encoding factual associations within over-parameterized foundation models. The prevailing paradigm posits that factual

knowledge is not distributed uniformly but resides in localized, directly-editable computations, particularly within the feed-forward networks (FFNs) of middle transformer layers [18]. This hypothesis is primarily validated through causal tracing techniques, such as representation denoising and activation patching, which intervene on neuron activations to quantify their decisive influence on factual predictions. By identifying components where activations mediate the recall of subject-predicate relations, researchers can target weight updates with surgical precision, thereby minimizing collateral damage to unrelated capabilities.

However, recent empirical inquiries have revealed a striking divergence between causal localization signals and optimal editing sites, challenging the assumption that mechanistic interpretability directly translates to editability. Counterintuitively, modifying weights in layers identified by causal tracing does not always yield superior editing performance compared to editing alternative layers [21]. This disconnect suggests that while certain neurons may be causally active during inference, they may not represent the most stable or efficient loci for parameter modification, potentially due to the complex superposition of features in deep representations. Furthermore, the efficacy of localization is highly sensitive to the specific editing objective; while localization correlates with success in amplifying facts, it offers limited guidance for erasing or overriding entrenched knowledge, indicating that the mechanisms for knowledge storage and retrieval are distinct from those governing plasticity.

The generalizability of these attribution methods faces significant hurdles when extending beyond standard Transformer architectures to emerging State Space Models (SSMs) like Mamba. Although SSMs share the goal of encoding sequential data into compressed latent spaces, their linear recurrence mechanisms differ fundamentally from the attention-based information flow of Transformers [80]. Preliminary analyses suggest that while factual storage in SSMs also concentrates in middle layers, the lack of explicit attention heads and the continuous-time dynamics of SSMs necessitate novel attribution frameworks that can disentangle the influence of recurrent states from static parameters [81]. Current causal tracing tools, designed for the discrete, token-wise attention masks of Transformers, may fail to capture the temporal dependencies and memory consolidation patterns inherent in SSMs, risking misidentification of editing targets in these efficient architectures.

Moreover, the reliance on component-level attribution must be reconciled with the reality of distributed representation, where multiple facts overlap in shared subspaces. Advanced component modeling algorithms, such as COAR, demonstrate that decomposing predictions into simple functional building blocks can enable diverse editing tasks, from fixing errors to forgetting specific classes [82]. Yet, the “localization-editing” gap implies that successful editing requires not only identifying where knowledge lives but also understanding the optimization landscape surrounding those parameters. Future research must therefore move beyond static attribution maps to develop dynamic, task-aware localization strategies that account for the interplay between architectural constraints, knowledge superposition, and the specific geometry of the loss landscape, ensuring that mechanistic insights robustly guide the next generation of lifelong adaptation systems.

4.2 Closed-Form and Constrained Parameter Updates

Building directly upon the mechanistic insights and localization strategies discussed previously, closed-form and constrained parameter updates represent the most direct mechanism for precise model editing. While the prior section highlighted the complexities in pinpointing optimal editing sites, this approach leverages mathematical optimization to inject new factual associations into foundation models once those sites are identified, aiming to minimize collateral damage to general

capabilities. These methods fundamentally treat the feed-forward networks (FFN) within transformer layers as key-value associative memories, where factual knowledge is stored in specific weight matrices that can be analytically updated. The seminal work [18] established this paradigm by introducing Rank-One Model Editing (ROME), which utilizes a closed-form solution to modify a single weight matrix such that it maps a specific subject key to a new object value, effectively rewriting the stored association without iterative gradient descent. This method relies on the assumption that the FFN layer operates as a linear associative memory, allowing for an exact solution that satisfies the edit constraint while minimizing the deviation from the original weights via a least-norm objective.

Building upon this foundation, recent advancements have sought to reconcile the tension between single-step precision and the scalability required for mass updates, addressing the limitations of modifying isolated layers noted in localization studies. The work [83] introduced MEMIT, which generalizes the rank-one update to accommodate thousands of simultaneous edits by distributing the update across multiple layers, thereby preventing the saturation of any single layer’s capacity. A critical theoretical unification of these locate-and-edit strategies is presented in [33], which posits that both ROME and MEMIT optimize a common “preservation-memorization” objective. This framework introduces the Equality-constrained Mass Model Editing (EMMET) algorithm, demonstrating that batched editing can be achieved through a closed-form equality constraint rather than the least-squares relaxation used in MEMIT, thus offering a symmetric approach to both single and batched scenarios.

However, the efficacy of direct weight modification remains contingent upon the precise isolation of components responsible for factual storage versus those governing general reasoning, a challenge underscored by the localization-editing gap. [84] argues that standard methods often inadvertently modify hidden states containing information from Multi-Head Self-Attention (MHSA) mechanisms, which encode general knowledge extraction patterns. By distinguishing between MHSA and FFN contributions, component-specific optimization ensures that weight updates target only the factual parameters, preserving the model’s linguistic coherence. Furthermore, the challenge of sequential editing remains a significant hurdle; naive application of closed-form updates often leads to “model collapse” or severe degradation in general abilities after a series of modifications, validating concerns about the stability of parameter modification raised earlier. [26] identifies that such failures are frequently artifacts of numerical instability and implementation irregularities rather than fundamental theoretical limits, proposing stabilized variants like r-ROME that sustain thousands of sequential edits.

Despite these algorithmic refinements, the trade-off between edit success and the preservation of general abilities remains precarious, suggesting that direct parameter manipulation may inherently disrupt the delicate superposition of features in over-parameterized regimes. Empirical analyses in [46] reveal that even state-of-the-art closed-form methods can drastically impair reasoning and question-answering capabilities. Consequently, while closed-form solutions offer unparalleled computational efficiency and theoretical elegance compared to fine-tuning [4; 85], their deployment in lifelong scenarios necessitates rigorous constraints on update magnitude and careful layer selection, as explored in lifelong editing frameworks like WilKE [11]. These inherent limitations in static weight manipulation motivate the paradigmatic shift toward externalized memory and retrieval-augmented architectures discussed in the following section, which seek to decouple knowledge storage from reasoning capabilities to better address the reliability and scalability challenges of lifelong adaptation. Future research must therefore focus on developing adaptive constraints that dynamically balance the fidelity of new knowledge injection against the stability of the model’s pre-trained

manifold, or alternatively, explore non-parametric avenues for enduring knowledge integration.

4.3 Externalized Memory and Retrieval-Augmented Architectures

Externalized memory and retrieval-augmented architectures represent a paradigmatic shift in model editing, moving away from the fragile manipulation of static weights toward decoupled systems where knowledge storage is distinct from reasoning capabilities. This architectural decoupling addresses the “impossible triangle” of reliability, generalization, and locality that plagues direct parameter updates in lifelong editing scenarios [9]. By externalizing updates to dynamic memory modules, these approaches enable scalable, self-updating systems capable of absorbing million-scale knowledge injections without degrading the base model’s pre-trained integrity [86].

A primary category of this approach involves integrating updatable memory pools directly within the model’s latent space. For instance, [86] introduces a fixed-size memory pool inside the transformer architecture, allowing the model to self-update with text knowledge and retain information over long horizons. This method contrasts sharply with traditional weight editing, as it avoids the numerical instability and condition number explosion associated with sequential matrix modifications [44]. Similarly, [10] proposes writing new factual mappings into discrete codebooks within the latent space. GRACE creates a local, discrete repository of edits that functions as a spot-fix mechanism for streaming errors, enabling thousands of sequential updates while leaving the underlying transformer weights untouched, thereby preserving general capabilities that are often compromised by direct fine-tuning [46].

Parallel to internal memory expansion, retrieval-augmented continuous prompting offers a mechanism to inject knowledge dynamically at inference time. The [78] framework exemplifies this by converting retrieved facts into continuous prompts that prefix the input query embeddings. This approach, known as RECIPE, utilizes a Knowledge Sentinel to dynamically threshold retrieval relevance, ensuring that the model refines its responses based on external knowledge without permanent parameter modification. This strategy effectively bridges the gap between static pre-training and dynamic world knowledge, mitigating the catastrophic forgetting observed in continual fine-tuning [7]. Furthermore, such retrieval-enhanced prompting facilitates multi-hop reasoning by iteratively refining answers based on retrieved context, a capability where purely parametric editors often fail due to a lack of consistent belief propagation [87].

The synergy between external memory and in-context learning further extends editing capabilities to black-box models. Approaches like [88] demonstrate that constructing demonstration contexts can achieve competitive edit success rates comparable to gradient-based methods like ROME, but with significantly reduced side effects on unrelated facts. This suggests that the context window itself can serve as a transient, high-bandwidth edit buffer. However, challenges remain in ensuring logical consistency across edits. While externalized memory improves locality, it often struggles with generalization, as retrieved facts may not be deeply integrated into the model’s reasoning pathways [9]. Future directions must therefore focus on hybrid architectures, such as the dual parametric memory schemes proposed in [9], which employ learned routers to direct queries to either pre-trained or edited subspaces, balancing the stability of the base model with the plasticity required for lifelong adaptation.

4.4 Activation Steering and Representation Tuning

Activation steering and representation tuning constitute a paradigm of non-invasive model editing that modulates foundation model behavior by intervening directly in the latent space, serving as a

critical bridge between the externalized memory architectures discussed previously and the holistic alignment strategies that follow. Unlike direct weight modification methods that risk destabilizing the broader knowledge distribution, or retrieval-based systems that rely on external buffers, these techniques operate by injecting steering vectors into residual streams or optimizing lightweight projection modules to shift activation trajectories toward desired semantic manifolds. This approach effectively decouples the reasoning capabilities encoded in the backbone from the behavioral policies required for specific tasks, offering a flexible mechanism for enhancing truthfulness, mitigating bias, and correcting hallucinations with minimal computational overhead.

A primary distinction within this domain lies between transient activation steering, which operates solely during inference, and permanent representation tuning, which embeds behavioral directions into the model via auxiliary parameters. Transient steering leverages the linearity of transformer representations to add a control vector $\mathbf{v}$ scaled by a coefficient α to the hidden states $\mathbf{h}_l$ at layer l , such that $\mathbf{h}'_l = \mathbf{h}_l + \alpha\mathbf{v}$. Recent advances in adversarial representation engineering demonstrate that such vectors can be derived to guide models toward complex conceptual attributes without degrading baseline performance, providing a unified framework for conceptual editing [89]. This method is particularly effective for addressing diverse hallucination categories, as it allows for adaptive shifts in output distributions toward factual domains without the need for retraining. Furthermore, the efficacy of steering relies on the modularity of internal representations; studies on configurable foundation models reveal that functional neuron partitions emerge during pre-training, allowing specific “bricks” of knowledge to be activated or suppressed dynamically [54].

Conversely, permanent representation tuning seeks to institutionalize these behavioral shifts through parameter-efficient fine-tuning (PEFT) mechanisms that avoid the plasticity loss associated with full model updates. Techniques such as Representation EDiting (RED) scale and bias layer-wise representations, achieving performance comparable to full fine-tuning while reducing trainable parameters by orders of magnitude [90]. Similarly, approaches utilizing low-rank adapters capture specific information along certain axes, enabling the modular adjustment of behaviors while preserving general utility [39]. The theoretical underpinning of these methods often involves optimizing dual loss functions, such as cosine similarity losses, to align internal representations with target vectors. This contrasts with methods that merely filter outputs, as representation tuning fundamentally alters the geometry of the latent space to make desired behaviors more accessible, laying the groundwork for the policy-level optimizations explored in subsequent sections.

However, the interplay between steering magnitude and model stability presents a critical trade-off that necessitates careful calibration. Excessive steering can push activations out-of-distribution, leading to coherence collapse or the amplification of other latent biases. Research into contrastive adapters suggests that bringing sample embeddings closer to class prototypes can significantly improve group robustness, yet improper adapter training may inadvertently increase accuracy gaps between subpopulations [91]. Moreover, while activation steering offers immediate corrigibility, it lacks the persistence of weight-based edits and the dynamic scalability of retrieval-augmented systems, necessitating the storage of steering vectors for every desired modification. Emerging hybrid architectures aim to resolve this by integrating steering mechanisms with retrieval-augmented systems, where retrieved facts are converted into continuous prompts to guide the model dynamically [6]. Future directions point toward neuro-symbolic integration, where steering vectors are derived from formal logical constraints rather than empirical data distributions, ensuring that activation modifications adhere to verifiable reasoning chains while maintaining the fluidity of neural adaptation required for the next generation of aligned foundation models.

4.5 Alignment-Based and Preference-Optimized Editing

Emerging frameworks increasingly reconceptualize knowledge editing not merely as a parameter-update problem, but as an alignment challenge where the objective is to refine model outputs to satisfy human preferences and factual correctness simultaneously. This paradigm shift leverages preference optimization techniques to ensure that edits align with human intent while preserving linguistic fluency and logical consistency. A pioneering approach in this domain is Knowledge Direct Preference Optimization (KDPO), which adapts the Direct Preference Optimization (DPO) loss structure for online editing. By treating current, potentially erroneous knowledge as negative samples and target facts as positive samples, KDPO enables refined, localized updates through teacher-forcing mechanisms, effectively bridging the gap between preference learning and factual correction [92]. This method contrasts with traditional locate-and-edit algorithms by optimizing the policy directly against a preference distribution rather than minimizing reconstruction error on specific tokens.

Complementing this, the Learning to Edit (LTE) framework introduces a two-phase process that explicitly aligns models on curated parallel datasets before deployment. In the alignment phase, models are fine-tuned to apply updated knowledge reliably while preserving out-of-scope information, effectively teaching the model “how to edit” rather than simply memorizing facts [93]. This is followed by a retrieval-based inference phase that ensures robustness in batch and sequential scenarios. Similarly, Reward rAnked FineTuning (RAFT) offers a streamlined alternative to Reinforcement Learning from Human Feedback (RLHF) by selecting high-quality samples based on a reward model and discarding those exhibiting undesired behavior, thereby enhancing model performance without the instability associated with standard RL algorithms [94]. These approaches collectively suggest that integrating preference signals during the editing process can mitigate the degradation of general abilities often observed in direct weight modification [46].

Furthermore, the integration of human feedback extends beyond textual facts to multimodal domains. The HIVE framework demonstrates that harnessing human feedback for instructional visual editing significantly improves adherence to user preferences by learning a reward function from human evaluations of edited images [95]. This highlights the versatility of preference-optimized editing across modalities. To address the logical consistency of edits, recent works propose verification-enhanced reasoning chains. For instance, DEEPEDIT treats knowledge editing as a constrained decoding problem, utilizing depth-first search to substitute new knowledge into reasoning steps, ensuring coherence between the edited fact and the derived conclusions [96]. This aligns with the broader goal of maintaining logical consistency in evolving models, as emphasized in frameworks that combine Chain-of-Thought reasoning with post-editing verification [93].

Despite these advancements, challenges remain in balancing specificity with generalization. While preference-optimized methods excel at aligning outputs with human intent, they must carefully manage the trade-off between enforcing new constraints and preserving the model’s inherent generative capabilities. Future directions likely involve hybrid architectures that combine the precision of parameter-efficient editing with the flexibility of preference optimization, potentially leveraging automated knowledge augmentation strategies to stabilize performance across sequential updates [97]. Ultimately, treating editing as an alignment problem provides a robust theoretical foundation for developing lifelong adaptation systems that are not only factually accurate but also socially aligned and logically coherent.

5 Synergies, Interplay, and Hybrid Adaptation Frameworks

5.1 Sequential Editing Dynamics and Interference Mitigation

Sequential model editing represents the critical frontier in transitioning foundation models from static repositories to dynamic, lifelong knowledge systems. However, naively applying single-edit algorithms sequentially often precipitates catastrophic performance degradation, a phenomenon rooted in the superposition of knowledge within over-parameterized weight spaces. Theoretical analysis reveals that as the number of edits increases, the interference term scales non-linearly due to the overlapping subspaces used to encode distinct facts, leading to the failure of closed-form solutions like ROME in long-horizon scenarios [22]. This “superposition bottleneck” suggests that without explicit orthogonality constraints or capacity expansion, the condition number of the edited matrices deteriorates, amplifying numerical sensitivity and perturbing general capabilities [44].

Empirical studies underscore the fragility of direct parameter modification under sequential loads. Early investigations into sequential editing revealed that specific “disabling edits” could induce immediate model collapse, rendering the system incoherent after only a few updates [26]. While stabilized implementations such as r-ROME have mitigated these implementation-specific artifacts, allowing for thousands of modifications, they do not fully resolve the fundamental trade-off between edit reliability and the preservation of pre-trained priors [26]. Comparative evaluations indicate that parameter-modifying methods consistently degrade fundamental reasoning and language modeling abilities as the edit sequence lengthens, whereas parameter-preserving approaches maintain general utility but often struggle with the accurate recall of edited knowledge in varied formats [98]. Furthermore, the accumulation of errors manifests not merely as forgetting but as “toxicity buildup” and “toxicity flashes,” where pattern mismatches in layer-wise updates cause sudden spikes in incoherent or harmful generations [11].

To mitigate these interference dynamics, recent frameworks have introduced rigorous mathematical constraints and architectural decoupling. The Perturbation Restraint on Upper bound for Editing (PRUNE) framework theoretically bounds the condition number of update matrices, demonstrating that restraining perturbations is essential for preserving general abilities during continuous adaptation [44]. Alternatively, architectural strategies like GRACE and WISE bypass weight superposition entirely by externalizing edits into discrete key-value adaptors or side memories, effectively sharding knowledge subspaces to prevent conflict between sequential modifications [10; 9]. These dual-memory schemes treat the base model as a frozen prior while routing queries through dynamic, editable modules, thereby isolating the interference term from the core representation. Additionally, batched editing approaches like EMMET and MALMEN offer a compromise, aggregating multiple updates into a single least-squares optimization to reduce the frequency of disruptive gradient steps, though they remain susceptible to capacity limits in indefinite streams [33; 99].

Future directions must address the “impossible triangle” of reliability, generalization, and locality in lifelong settings. While current methods like WilKE improve layer selection via pattern matching to reduce toxicity, a unified theory of lifelong editing requires integrating continual learning regularization—such as elastic weight consolidation adapted for edit traces—with retrieval-augmented generation to ensure logical consistency across infinite horizons [11; 78]. Ultimately, robust sequential editing demands a shift from viewing edits as isolated weight perturbations to managing them as evolving states within a structured, interference-aware memory architecture.

5.2 Architectural Decoupling and Dual-Memory Hybrid Systems

Building upon the theoretical limits of sequential weight modification and the “superposition bottleneck” identified previously, architectural decoupling emerges as a structural solution to the “impossible triangle” of reliability, generalization, and locality. Rather than attempting to optimize conflicting objectives within a monolithic weight space, this paradigm strictly segregates static pre-trained knowledge from dynamic updates. The base model serves as an immutable repository of general capabilities, while plasticity is confined to auxiliary structures, thereby preventing the corruption of foundational representations during sequential modifications. A seminal realization of this philosophy is the dual parametric memory scheme, exemplified by [9], which introduces a side memory dedicated exclusively to edited knowledge alongside the main pre-trained memory. By employing a learned router to direct queries to the appropriate memory subspace, such systems effectively isolate edit interference, ensuring that localized updates do not propagate distal errors to unrelated knowledge domains.

Complementing parametric decoupling, discrete key-value adaptors offer a mechanism for high-frequency, streaming edits without altering underlying transformer weights. The [10] framework demonstrates this by writing new factual mappings into a discrete codebook within the model’s latent space. This method enables thousands of sequential spot-fixes with minimal impact on unrelated inputs, effectively bypassing the superposition bottleneck that plagues direct weight modification techniques like ROME or MEMIT in long-horizon scenarios. Similarly, [86] integrates a fixed-size, self-updatable memory pool directly into the transformer’s latent architecture, allowing for million-scale knowledge injections while maintaining operational integrity. These memory-augmented approaches contrast sharply with traditional locate-and-edit methods, shifting the burden of retention from weight optimization to explicit memory retrieval and modulation.

Beyond explicit memory banks, subspace isolation techniques provide a geometric solution to conflict resolution by allocating distinct parameter subspaces for different sets of edits, allowing diverse knowledge updates to merge without interference. This aligns with the broader concept of configurable foundation models, where functional “bricks” or neuron partitions are dynamically retrieved, merged, or grown to accommodate new capabilities [54]. Such modular operations allow the model to expand its capacity selectively, mirroring the dynamic expandability seen in lifelong learning networks like DEN [100] but adapted for the precision requirements of knowledge editing. Furthermore, retrieval-augmented interfaces externalize knowledge updates entirely, utilizing in-context learning or constrained decoding to inject facts at inference time, as seen in approaches that convert retrieved facts into continuous prompts [93].

Despite their promise in resolving interference, these hybrid systems introduce new challenges regarding router training stability and the latency overhead of multi-path inference. The efficacy of the decoupling strategy relies heavily on the router’s ability to discern context switches accurately; failure modes often manifest as “memory leakage,” where the model incorrectly accesses the side memory for general queries or vice versa. Moreover, while decoupling isolates edits, it does not inherently guarantee that these updates remain invariant under subsequent sequential training streams. Future directions must therefore address the synchronization between fast, discrete memory updates and the slow consolidation of knowledge into the base model, potentially leveraging bio-inspired complementary learning systems [101] to bridge the gap between transient editing and permanent structural adaptation. This necessity for robust consolidation mechanisms to protect edited parameters during lifelong adaptation sets the stage for the discussion on regularization and replay strategies in the following section.

5.3 Consolidation Mechanisms for Edit Preservation

The integration of model editing into lifelong learning pipelines necessitates robust consolidation mechanisms to ensure that precise, localized updates remain invariant under subsequent sequential training. While traditional model editing algorithms like [18] and [83] demonstrate efficacy in static settings, they often suffer from rapid degradation when subjected to continual fine-tuning streams, a phenomenon exacerbated by the “impossible triangle” of reliability, generalization, and locality in lifelong scenarios [9]. To bridge this gap, recent research adapts continual learning regularization strategies to explicitly protect edited parameters, treating them as high-importance knowledge that must be preserved against catastrophic interference.

A primary approach involves modifying Fisher-based regularization techniques, such as Elastic Weight Consolidation (EWC), to assign disproportionately high importance scores to weights altered during the editing phase. By penalizing deviations from these edited states during subsequent task optimization, the model is constrained to find solutions in the null space of previous edits, akin to the principles established in [69] and [102]. However, standard EWC may be insufficient for the non-convex landscapes of foundation models; thus, advanced variants employ asymmetric constraints that strictly forbid overwriting edited components while allowing plasticity in untouched subspaces. This aligns with findings from [103], which suggests that sparse masking and sensitivity analysis can identify critical “model patches” that require decoration rather than replacement to maintain both old and new task performance.

Beyond parameter penalization, replay-based consolidation has evolved to specifically reinforce edited facts. Instead of generic experience replay, methods now utilize generative models to synthesize counterfactual examples that target the specific knowledge triplets updated during editing. [58] established the viability of generative rehearsal, yet modern adaptations like [104] refine this by editing stored exemplars to maximize loss on the current model state, thereby creating “challenging” replay buffers that specifically harden the model against forgetting recent corrections. Furthermore, [93] proposes a dual-phase framework where alignment on curated parallel datasets ensures that the model learns to apply updated knowledge robustly, effectively internalizing the edit as a permanent behavioral trait rather than a fragile weight perturbation.

Emerging hybrid architectures address the limitations of weight-based consolidation by decoupling storage from computation. [9] introduces a dual parametric memory scheme, segregating pre-trained knowledge from edited knowledge into distinct subspaces merged via a learned router, thereby preventing direct conflict. Similarly, [10] bypasses weight modification entirely, writing edits into a discrete codebook within the latent space, which inherently protects edits from gradient-based erosion during subsequent training. Despite these advances, challenges remain in scaling these mechanisms to infinite horizons without accumulating toxicity or logical inconsistencies, as noted in [11]. Future directions must therefore focus on dynamic, context-aware consolidation schedules that balance the rigidity required for edit preservation with the plasticity necessary for absorbing novel domain shifts, potentially leveraging neuro-symbolic constraints to enforce logical consistency across sequential modifications.

5.4 Conflict Resolution and Logical Consistency in Evolving Models

Building upon the consolidation mechanisms designed to preserve edited parameters against catastrophic forgetting, the next critical challenge lies in maintaining the logical integrity of evolving foundation models amidst sequential modifications. As edits accumulate, models face the risk of “toxicity buildup” and “toxicity flash,” where successive updates degrade global coherence or in-

duce sudden performance collapses due to pattern mismatches between new knowledge and ill-suited layer representations [11]. To mitigate these interference effects, advanced strategies employ layer-wise conflict detection, selecting editing sites based on pattern matching degrees rather than static localization heuristics, thereby preserving the orthogonality between new and existing knowledge subspaces [11].

Beyond precise layer selection, ensuring logical consistency requires propagating edits to entailed facts to prevent knowledge fragmentation. Standard editing methods often fail to capture these “ripple effects,” resulting in models that recall an updated fact but fail to adjust related beliefs, leading to logical incoherence [19; 87]. To address this, graph-enhanced approaches leverage external knowledge structures to identify associated concepts, ensuring that an update to a single entity automatically triggers consistent modifications across the relevant semantic neighborhood [105]. This aligns with the broader philosophical challenge of belief revision, where the goal shifts from mere memorization to the maintenance of a probabilistically coherent world model [49].

Furthermore, the numerical stability of sequential edits is paramount for long-term consistency. Theoretical analyses indicate that the condition number of edited weight matrices serves as a proxy for perturbation sensitivity; as the number of edits increases, unchecked growth in the condition number exacerbates the degradation of general capabilities [44]. Frameworks like PRUNE explicitly constrain this condition number to lower the upper bound of perturbation, effectively balancing edit success with the preservation of the model’s original reasoning abilities. Complementing this, dual-memory architectures circumvent direct parameter conflicts by sharding knowledge into distinct subspaces or side memories, utilizing routers to dynamically access the most relevant knowledge state without overwriting the foundational backbone [9; 10].

Emerging frontiers also explore the integration of unlearning mechanisms to actively remove outdated or contradictory information before injecting new facts, treating continuity as a joint optimization of acquisition and forgetting [106]. Additionally, verification-enhanced reasoning chains and constrained decoding strategies offer inference-time safeguards, correcting logical inconsistencies in generated paths before they manifest as output errors [96; 93]. Ultimately, achieving robust logical consistency in evolving models demands a shift from isolated edit operations to holistic frameworks that jointly optimize for locality, generalization, and semantic coherence across infinite horizons.

6 Evaluation Protocols, Benchmarks, and Application Domains

6.1 Quantitative Metrics for the Stability-Plasticity-Locality Trilemma

Evaluating the efficacy of lifelong adaptation in foundation models requires moving beyond static accuracy metrics to a multidimensional framework that quantifies the intricate trade-offs within the Stability-Plasticity-Locality trilemma. In continual learning, stability is traditionally measured by average forgetting (F_t) and backward transfer, capturing the degradation of performance on previous tasks as new data streams are ingested [3]. However, for foundation models, these metrics must be augmented with representation forgetting measures, which assess the geometric distortion of feature spaces independent of classifier heads [43]. Plasticity, conversely, is not merely the accuracy on current tasks but the efficiency of adaptation under computational constraints; recent work emphasizes “computationally budgeted” metrics that evaluate learning speed relative to FLOPs or wall-clock time, revealing that many rehearsal-based methods fail when compute, rather than memory, is the bottleneck [107]. Furthermore, the phenomenon of plasticity decay—the loss of a model’s ability to learn over infinite horizons—necessitates dynamic metrics tracking recovery rates

after regularization or reset interventions [35].

In the domain of model editing, the evaluation paradigm shifts toward the triad of Reliability, Locality, and Portability (or Generalization). Reliability, often termed Edit Success Rate (ESR), quantifies the probability that an edited model outputs the target fact for the specific edited input [18]. Locality measures the preservation of unrelated knowledge, ensuring that the edit does not induce “ripple effects” or catastrophic interference on distal facts; this is critical, as naive editing often degrades general capabilities despite high success rates on target queries [46]. Advanced specificity metrics now employ KL-divergence to detect subtle distributional shifts in non-edited contexts, offering a more sensitive probe than binary accuracy [108]. Portability evaluates whether the edited knowledge generalizes to paraphrased inputs and multi-hop reasoning chains, a dimension where current methods frequently fail, as evidenced by benchmarks like MQuAKE which test entailed consequences of edits rather than simple recall [87].

The synthesis of these fields reveals a fundamental tension: methods optimizing for high plasticity or edit success often compromise stability and locality. For instance, while gradient-based editing offers high reliability, it frequently suffers from poor locality in sequential settings due to knowledge superposition [83]. Conversely, retrieval-augmented approaches excel in locality by isolating updates in external memory but often struggle with portability and deep reasoning integration [6]. To address this, emerging metrics focus on the “impossible triangle” of lifelong editing, quantifying the trade-off between reliability, generalization, and locality over thousands of sequential updates [9]. Future evaluation protocols must therefore adopt longitudinal metrics that track the accumulation of toxicity and logical inconsistencies over extended adaptation cycles, ensuring that foundation models remain robust not just after a single edit, but throughout a lifetime of evolution [11].

6.2 Benchmarking Frameworks for Non-Stationary and Dynamic Streams

Building upon the multidimensional metrics and trade-offs defined in the Stability-Plasticity-Locality trilemma, the practical assessment of continual learning and model editing necessitates a paradigm shift from static, offline benchmarks to dynamic frameworks that faithfully simulate non-stationary data streams and evolving knowledge landscapes. Traditional evaluation protocols, often relying on artificially permuted datasets like Split-CIFAR or Permuted-MNIST, fail to capture the natural temporal evolution and semantic drift characteristic of real-world deployments [3; 12]. To address this disconnect, recent initiatives have introduced benchmarks designed to test stability and plasticity under realistic conditions. For instance, CORE50 established an early standard for continuous object recognition by providing a dataset with natural temporal variation in robotic vision scenarios, moving beyond disjoint task boundaries [109]. Similarly, the CLEAR benchmark leverages a decade of real-world imagery (2004–2014) to evaluate continual learning under smooth concept evolution, proposing “streaming” protocols that test on future data to prevent the performance inflation associated with i.i.d. assumptions [110].

In the domain of language foundation models, the evaluation challenge shifts toward balancing the retention of time-invariant knowledge with the acquisition of updates. TemporalWiki addresses this by utilizing consecutive snapshots of Wikipedia and Wikidata to construct a lifelong benchmark, enabling the periodic assessment of a model’s ability to adapt to frequently updated corpora without catastrophic forgetting [111]. Complementary to this, the Continual Knowledge Learning (CKL) framework formulates specific metrics to quantify the triad of knowledge retention, outdated knowledge updating, and new knowledge acquisition, highlighting that parameter expansion is often requisite for reliable performance in these dynamic settings [112]. Furthermore, for multimodal

systems, CLiMB provides a critical testbed for vision-and-language tasks, revealing that while standard continual learning methods mitigate forgetting, they often fail to facilitate the cross-task knowledge transfer essential for general-purpose assistants [113].

Critically, aligning with the reliability, locality, and portability triad discussed previously, the evaluation of model editing requires benchmarks that extend beyond simple fact recall to assess logical consistency and ripple effects. The MQuAKE benchmark introduces multi-hop questions to determine if an edit to a specific fact (e.g., a political leader) correctly propagates to entailed queries (e.g., their spouse), exposing the catastrophic failure of current editors to maintain coherent knowledge graphs [87]. Expanding on this, RippleEdits constructs diagnostic scenarios to evaluate whether editing one fact induces necessary updates to related attributes, such as familial relationships, thereby testing the locality and portability of edits in a rigorous manner [19]. Additionally, specialized suites like those proposed in [16] and [27] offer standardized datasets for counterfactual assertions and dialogue generation, facilitating robust comparisons between locate-and-edit methods and retrieval-augmented approaches.

Despite these advancements, significant gaps remain that hinder the transition to the high-stakes applications discussed next. Many existing benchmarks still operate under compute-unconstrained assumptions, which do not reflect the strict latency and memory budgets of edge deployment [107]. Moreover, the scarcity of benchmarks for sequential model editing (SME) that simulate thousands of continuous corrections limits the understanding of long-term editing integrity [22]. Future frameworks must integrate adversarial robustness testing within these dynamic streams, as non-stationary environments introduce unique vulnerabilities to poisoning and distribution shifts that static evaluations cannot detect [114; 115]. Ultimately, the field requires unified simulation environments that concurrently stress-test computational efficiency, logical consistency, and adaptability across multimodal, time-varying distributions to ensure safe deployment in domains such as healthcare and robotics.

6.3 Domain-Specific Deployment and Application Case Studies

The deployment of foundation models in high-stakes domains necessitates a paradigm shift from static inference to dynamic, lifelong adaptation, where continual learning (CL) and model editing serve as critical mechanisms for maintaining safety, relevance, and operational efficiency. In healthcare, the volatility of medical knowledge and the scarcity of labeled data render static pre-training insufficient. Foundation models in this sector must integrate new clinical guidelines and patient-specific distributions without compromising diagnostic accuracy. Approaches leveraging continual pretraining allow models to specialize in emerging domains while retaining general biomedical knowledge, yet they face the risk of catastrophic forgetting when domain sequences exhibit low semantic similarity [116]. To mitigate this, data-centric strategies emphasize rigorous data characterization and quality control during the adaptation phase, ensuring that updates align with human values and privacy constraints [117]. Furthermore, precise model editing offers a pathway for correcting specific diagnostic errors or updating drug interaction facts without full retraining, addressing the “belief revision” challenge where rational updates must preserve logical consistency across the knowledge graph [49].

In autonomous robotics and cyber-physical systems, the requirement for real-time adaptation to non-stationary environments is paramount. Traditional policies often fail when encountering novel objects or morphological changes post-deployment. Recent frameworks demonstrate that adapting pre-trained policies via off-policy reinforcement learning can achieve substantial performance gains

with minimal data, effectively enabling robots to learn from mistakes continuously [118]. However, the integration of foundation models into the robot autonomy stack introduces challenges in safety guarantees and uncertainty quantification, particularly when leveraging large language models for high-level planning [119]. The fusion of CL with digital twin technologies further enhances this capability; by employing elastic weight consolidation at the network edge, cyber twins can synchronize with physical counterparts in dynamically changing environments, minimizing de-synchronization gaps while preventing the erosion of historical driving policies [120].

The domain of Earth observation and climate monitoring presents unique challenges due to the massive scale of streaming geospatial data and the multi-modal nature of satellite imagery. Foundation models like Prithvi have demonstrated efficacy in fine-tuning for tasks such as flood mapping and wildfire segmentation, yet they require continual adaptation to reflect rapidly changing land cover and climate patterns [121]. The development of neural plasticity-inspired architectures, such as Dynamic One-For-All (DOFA), allows for the adaptive integration of diverse sensor modalities, enabling unified analysis across optical, radar, and hyperspectral data streams [122]. Despite these advances, the scarcity of annotated data remains a bottleneck, necessitating self-supervised continual learning protocols that can operate effectively on unlabelled data streams [123].

Across these domains, a common tension exists between the stability of pre-trained representations and the plasticity required for domain-specific evolution. While parameter-efficient fine-tuning and modular “brick” operations offer scalable solutions for customizing models without full retraining [54], the long-term reliability of sequentially edited models remains an open challenge. Future deployment frameworks must therefore prioritize hybrid architectures that decouple static core knowledge from dynamic, editable memory modules, ensuring that local corrections do not induce global capability collapse. This synthesis of continual adaptation and precise editing is not merely an optimization but a fundamental prerequisite for the safe and effective deployment of foundation models in our ever-evolving world.

6.4 Robustness, Security, and Reliability Assessment Protocols

Building on the deployment challenges and hybrid architectures outlined previously, the assessment of robustness, security, and reliability in dynamically evolving foundation models necessitates a paradigm shift from static evaluation to continuous, lifecycle-aware protocols. Unlike stationary models, continually learning and edited systems face unique vulnerabilities where the very mechanisms designed for adaptation become primary attack surfaces. A primary concern is the susceptibility of sequential learning streams to data poisoning, where adversaries inject malicious samples to induce targeted forgetting or cross-task contamination with minimal detectable footprint [124]. This risk is exacerbated in model editing contexts, where lightweight modification techniques can be exploited to inject persistent backdoors using as few as 15 samples, compromising model integrity while maintaining benign performance on standard benchmarks [125]. Consequently, evaluation protocols must rigorously test adaptation phases against such adversarial exploits, moving beyond simple accuracy metrics to measure the resilience of parameter-efficient modules and editing algorithms against gradient manipulation and unauthorized modification [115].

Reliability assessment further demands sophisticated monitoring for error propagation and model collapse, phenomena where imperfect edits or noisy data streams compound over time to degrade global capabilities—a direct consequence of the “global capability collapse” risks highlighted in prior deployment scenarios. Theoretical analyses indicate that sequential editing can exacerbate numerical sensitivity, where the condition number of edited matrices grows with each update, leading to a

deterioration of general abilities [44]. Empirical evidence suggests that without proper constraints, disabling edits can cause immediate model collapse, necessitating stabilized implementations and perturbation-restrained frameworks to ensure longevity [26]. Moreover, the “stability gap” phenomenon reveals that models may suffer substantial temporary forgetting upon encountering new tasks, a failure mode often masked by standard end-of-task evaluations but critical for real-time reliability [126]. Protocols must therefore employ per-iteration evaluation to quantify worst-case performance dips and detect the accumulation of errors that lead to systemic failure [127].

Auditability and provenance tracking represent another critical pillar of reliability, addressing the difficulty of verifying continuously changing model states, particularly within the decentralized or federated adaptation frameworks required for edge and robotic applications. As models evolve through these distributed pathways, maintaining an auditable trail of modifications is essential for accountability and regulatory compliance [128]. Emerging frameworks propose leveraging distributed ledgers and version control systems to track model lineage, ensuring that every edit or update can be traced to its data source and rationale [1]. Furthermore, the evaluation of specificity must extend beyond immediate locality to detect unwanted side effects and factual drift in long-form generation, as current metrics often fail to capture the subtle degradation of coherence and consistency over extended interaction sequences [129; 108]. Future assessment protocols must integrate these dimensions, combining adversarial stress testing, numerical stability analysis, and provenance verification to create a holistic safety framework. Only through such comprehensive evaluation can we ensure that foundation models remain trustworthy as they transition from static artifacts to autonomous, self-evolving agents capable of operating safely in open-world environments.

7 Open Challenges, Ethical Considerations, and Future Directions

7.1 Theoretical Foundations for Infinite-Horizon Adaptation

The transition from finite-task continual learning to infinite-horizon adaptation presents a fundamental theoretical crisis: the absence of provable convergence guarantees for over-parameterized foundation models subjected to unbounded sequences of updates. While classical analyses in linear regimes suggest that forgetting bounds scale with task similarity and ordering [34], these insights fail to capture the complex dynamics of deep, non-convex landscapes where task boundaries are often ambiguous [130]. Current empirical evidence indicates that optimal continual learning theoretically requires perfect memory, rendering the problem NP-hard in general settings [30]. Consequently, practical algorithms rely on approximations that lack rigorous stability bounds over infinite time horizons, leaving models vulnerable to gradual capability erosion.

A central theoretical obstruction is the “impossibility triangle” of lifelong editing, which posits that reliability, generalization, and locality cannot be simultaneously optimized during sequential modifications [9]. This trilemma arises from the phenomenon of knowledge superposition, where multiple facts are encoded in overlapping parameter subspaces. When edits are applied sequentially, interference terms accumulate, leading to a degradation of general abilities that scales with the number of updates [46]. Recent theoretical work identifies the condition number of edited matrices as a critical factor; as sequential edits increase numerical sensitivity, the perturbation upper bound expands, exacerbating the deterioration of pre-trained knowledge [44]. This suggests that without explicit constraints on the spectral properties of weight updates, infinite-horizon adaptation is mathematically destined to converge toward a degenerate solution.

Furthermore, the dynamics of error propagation in infinite streams reveal a “stability gap,” where models experience temporary but severe performance drops upon encountering new distributions

before recovering [126]. In over-parameterized regimes, this is compounded by the loss of plasticity, where the network’s ability to learn new concepts diminishes as parameters saturate or align too closely with historical gradients [131]. Theoretical frameworks attempting to resolve this via function-space variational inference offer promising avenues by regularizing predictive distributions rather than weights, yet they struggle to scale to the dimensionality of modern foundation models [132]. Similarly, while architectural solutions like dynamic expert expansion [37] or discrete key-value adaptors [10] empirically mitigate interference, they lack formal proofs guaranteeing bounded regret over infinite horizons.

Future theoretical directions must move beyond static task-incremental assumptions to develop adaptive control theories for time-varying feature distributions. This includes deriving tighter generalization bounds for parameter-efficient tuning methods under non-stationary streams [5] and establishing conditions under which retrieval-augmented mechanisms can mathematically decouple working memory from long-term parameter storage to break the impossibility triangle. Ultimately, achieving safe infinite-horizon adaptation requires a shift from heuristic regularization to principled, geometry-aware optimization constraints that formally bound the accumulation of interference in high-dimensional manifolds.

7.2 Security Vulnerabilities in Dynamic and Collaborative Ecosystems

Bridging the theoretical fragility of infinite-horizon adaptation with practical security realities, the transition from static pre-training to dynamic, lifelong paradigms fundamentally expands the attack surface of foundation models. While the previous section highlighted the mathematical inevitability of interference and the “impossibility triangle” in sequential updates, these theoretical vulnerabilities manifest concretely as unique attack vectors inherent to continuous data streams and decentralized modification mechanisms. Unlike traditional static models where security is assessed at a single deployment point, dynamically evolving systems face persistent threats where adversaries exploit the very mechanisms designed for plasticity—such as experience replay, parameter-efficient fine-tuning, and model editing—to compromise model integrity over time. A primary concern is the susceptibility of continual learning streams to sophisticated poisoning attacks. In scenarios involving blurry task boundaries or non-stationary data distributions, adversaries can inject malicious samples that induce targeted forgetting or cross-task contamination, eroding specific capabilities with a minimal detectable footprint [133]. The reliance on episodic memory for replay-based consolidation further exacerbates this risk; if the diversity and purity of stored exemplars are not rigorously managed, poisoned anchors can propagate errors across sequential tasks, leading to systemic degradation that standard regularization constraints fail to mitigate [134].

Furthermore, the emergence of efficient model editing techniques has unveiled a potent vector for backdoor injection that bypasses the data volume requirements of traditional poisoning, directly exploiting the spectral sensitivities discussed earlier. Recent frameworks demonstrate that backdoors can be embedded directly into large language models through lightweight knowledge editing, altering a sparse subset of parameters to incorporate malicious triggers while maintaining benign performance on standard inputs [125]. This “surgical” compromise poses a critical auditing challenge: as models undergo thousands of sequential edits to correct facts or update knowledge, distinguishing between legitimate modifications and adversarial perturbations becomes increasingly difficult. The phenomenon of “disabling edits,” where specific modifications cause immediate model collapse, highlights the fragility of current editing algorithms, yet stabilized variants capable of sustaining massive sequential updates also risk silently accumulating adversarial biases if not properly constrained [26]. This accumulation of unverified changes creates the opaque, non-deterministic tra-

jectories that complicate accountability, setting the stage for the governance challenges addressed in the subsequent section.

In collaborative and federated ecosystems, these risks are compounded by the heterogeneity of data sources and the decentralization of training. The integration of foundation models into federated learning frameworks introduces supply chain vulnerabilities where compromised primitive models or malicious clients can propagate defects across diverse downstream applications [135]. While parameter-efficient adaptation methods like Low-Rank Adaptation (LoRA) offer computational efficiency, they may inadvertently create security gaps; lightweight adapter modules can be susceptible to gradient manipulation or unauthorized modification without triggering the defense mechanisms designed for full-model fine-tuning [39]. Moreover, the use of synthetic data for generative replay, while addressing privacy constraints, opens the door to “model collapse” feedback loops where errors or biases in generated data accumulate over iterations, progressively degrading the model’s distributional fidelity [66]. Addressing these vulnerabilities requires a paradigm shift from static robustness verification to dynamic, provenance-aware security protocols that can audit the lineage of every weight update and detect anomalous adaptation trajectories in real-time. Future research must prioritize the development of “self-destructing” mechanisms or task-blocking paradigms that prevent the repurposing of adapted models for harmful objectives without sacrificing their capacity for legitimate lifelong learning [136], thereby providing the technical safeguards necessary for the socio-technical governance frameworks required to manage these dynamic systems.

7.3 Ethical Governance, Accountability, and Participatory Frameworks

The governance of foundation models undergoing continual adaptation and precise editing presents a profound socio-technical challenge, fundamentally disrupting traditional paradigms of liability and accountability. As models transition from static artifacts to dynamic systems capable of autonomous self-evolution or decentralized modification, a “responsibility gap” emerges where harms caused by updated knowledge states cannot be traced to a single human authorizer. This is particularly acute in scenarios where models learn from deployment data streams, as seen in conversational agents designed for lifelong interaction [137]. The core difficulty lies in auditing time-varying model states; unlike static deployments, continually learning systems possess a non-deterministic trajectory where the provenance of specific behaviors becomes obscured by sequential updates. To address this, recent proposals advocate for decentralized governance architectures leveraging blockchain and specialized version control systems, such as Git-Theta, to maintain an immutable ledger of model lineage and enforce access policies across distributed contributions [138].

Furthermore, the interplay between continual learning and bias mitigation introduces complex feedback loops. When models adapt to non-stationary data distributions without rigorous oversight, they risk amplifying historical biases or entrenching unfair patterns through “algorithmic reparation” failures. This risk is exacerbated in federated settings where heterogeneous data distributions across clients can lead to concept drift that disproportionately affects marginalized groups [139]. The integration of human feedback into editing processes, while promising for alignment, also carries the danger of encoding subjective preferences that may not generalize equitably [95]. Consequently, governance frameworks must move beyond post-hoc auditing to incorporate participatory layers that allow diverse stakeholders to influence adaptation at both the infrastructure (“subfloor”) and application (“surface”) levels. Such participatory mechanisms are essential to prevent the centralization of power and ensure that local contextual needs are met without compromising global safety standards [1].

A critical technical dimension of ethical governance involves the implementation of “self-destructing” mechanisms or task-blocking paradigms that impede adaptation to harmful domains while preserving utility for beneficial tasks [136]. In the context of model editing, this requires formalizing constraints that prevent the injection of toxic knowledge or the erosion of safety boundaries during sequential updates. However, current methods often lack the theoretical guarantees to ensure that such constraints hold over infinite horizons, leading to potential “toxicity buildup” where minor adversarial edits accumulate into systemic failures [49]. Future research must therefore prioritize the development of verifiable adaptation protocols that combine cryptographic provenance tracking with real-time consistency checks. Only by integrating these technical safeguards with robust legal frameworks can we bridge the gap between the rapid pace of algorithmic evolution and the slower mechanisms of societal accountability, ensuring that lifelong learning systems remain aligned with human values throughout their operational lifecycle.

7.4 Emerging Frontiers in Architecture, Integration, and Deployment

Building on the imperative for verifiable adaptation protocols and robust safeguards outlined previously, the trajectory of lifelong adaptation for foundation models is shifting from monolithic fine-tuning toward modular, neuro-symbolic, and bio-inspired architectures that prioritize logical consistency and deployment efficiency. A primary frontier involves the integration of symbolic reasoning engines with statistical foundation models to mitigate the hallucination risks inherent in purely data-driven continual learning. By embedding verifiable constraints and knowledge graphs directly into the adaptation loop, future systems can ensure that sequential edits maintain global logical coherence, directly addressing the “toxicity buildup” and lack of theoretical guarantees observed in naive weight modification strategies [140]. This neuro-symbolic synergy is particularly critical for scientific domains, where models like [141] demonstrate that synergizing human-designed algorithms with data-driven pre-training yields high-accuracy simulations of dynamical systems that pure neural approaches cannot match.

Parallel to establishing logical rigor, the architectural paradigm is evolving toward “configurable foundation models” that treat neural networks as assemblies of functional “bricks” rather than static blocks. This modular perspective enables dynamic operations such as retrieval, merging, and growing of specific neuron partitions, allowing models to adapt to new tasks by activating sparse sub-networks rather than updating all parameters [54]. Such modularity is essential for scaling lifelong learning, as it permits the isolation of new knowledge in dedicated adapters or memory pools, thereby preserving the integrity of pre-trained representations and facilitating the precise, localized edits required for safe governance. Complementing this, bio-inspired mechanisms drawing from Complementary Learning Systems (CLS) theory propose dual-memory architectures where a fast system captures specific experiences and a slow system consolidates structured knowledge [142]. Recent advances in neural plasticity, such as the [57] architecture, further refine this by employing sparse networks with connection rewiring to create stable paths for old tasks while maintaining plastic paths for new ones, significantly reducing parameter overhead and enhancing resilience against catastrophic forgetting.

To realize these advanced architectures in practice, deployment scalability demands a radical rethinking of adaptation for resource-constrained edge environments. The prevailing cloud-centric paradigm is unsustainable for real-time applications in robotics and IoT, necessitating on-device continual learning frameworks that operate within strict memory and energy budgets. Innovations in quantized latent replays and specialized hardware, such as the PULP processor platform, have demonstrated that continual learning can be achieved on microcontrollers with minimal accuracy

loss [60]. Furthermore, efficient adaptation strategies like [143] enable privacy-preserving transfer learning without accessing full model weights, a crucial feature for federated ecosystems where data silos and regulatory constraints prevent centralization [135; 144]. The convergence of these trends points toward “foundation agents” capable of autonomous self-improvement through interaction with dynamic environments, necessitating new evaluation metrics for agency and value alignment [145]. Ultimately, the future of lifelong adaptation lies in hybrid systems that combine the generalization power of foundation models with the precision of symbolic logic, the efficiency of modular architectures, and the sustainability of edge-native learning protocols.

8 Conclusion

This subsection synthesizes the critical insights gathered throughout our survey, underscoring that the transition from static pre-training to dynamic adaptation is not merely an incremental improvement but a fundamental prerequisite for the next generation of foundation models. We have established that continual learning and model editing, historically treated as disjoint silos, must be unified into a cohesive lifelong adaptation paradigm to address the Stability-Plasticity-Locality trilemma. Methodological advancements have largely bifurcated into parameter-efficient isolation strategies and direct weight manipulation techniques. On one hand, approaches utilizing low-rank adapters and modular expansions, such as those explored in [5] and [53], successfully decouple new task acquisition from core representations, thereby mitigating catastrophic forgetting. On the other hand, precise editing algorithms like [83] and [84] demonstrate that targeted weight updates can correct factual errors with minimal computational overhead. However, our comparative analysis reveals a critical limitation: while parameter-efficient methods excel in plasticity, they often struggle with the seamless integration of dense, interconnected knowledge, whereas direct editing methods frequently succumb to interference and model collapse during sequential operations, as evidenced by the instability issues addressed in [26].

The survey further highlights that the “impossible triangle” of reliability, generalization, and locality in lifelong editing remains a significant theoretical bottleneck. Recent hybrid architectures, such as the dual-memory schemes proposed in [9] and the discrete key-value adapters in [10], offer a promising resolution by externalizing dynamic knowledge into side memories or latent codebooks. These approaches effectively bypass the superposition limits inherent in over-parameterized weights, allowing for thousands of sequential edits without degrading the base model’s utility. Furthermore, the integration of retrieval-augmented mechanisms, as seen in [6] and [93], suggests a paradigm shift where the model acts as a reasoning engine over a mutable knowledge store rather than a static repository of facts. This decoupling is essential for maintaining logical consistency and preventing the toxicity buildup observed in naive sequential editing scenarios [11].

Looking forward, the path toward truly adaptive and trustworthy artificial intelligence requires rigorous evaluation protocols that move beyond static benchmarks to simulate non-stationary, open-world streams. As argued in [146], current metrics often fail to capture the nuances of representation drift and long-term interference. Future research must prioritize the development of theoretical bounds for infinite-horizon adaptation, ensuring that convergence and stability are mathematically guaranteed even under indefinite update cycles. Moreover, the convergence of neuro-symbolic reasoning with continuous adaptation, alongside bio-inspired consolidation mechanisms, presents a fertile ground for overcoming the current limits of plasticity decay. Ultimately, the synthesis of continual learning and model editing will enable foundation models to evolve autonomously, correcting their own hallucinations and adapting to shifting human values in real-time, thereby realizing the vision of self-evolving systems that are both robust and aligned with societal needs.

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